CO1	Define various Object Oriented concepts in Java Programming.
CO2	Compare different data types in java.
CO3	Differentiate between built-in and user defined functions/methods, interfaces and packages etc.
CO4	Outline the importance of exception handling in programs.
CO5	Explain advanced concepts like multithreading, applet used in java.

Detailed Contents	Contact hours
Unit-I **Java Programming Fundamentals:** Introduction to Java, Stage for Java, Origin, Challenges of Java, Java Features, Java Program Development, Object Oriented Programming. [CO2] **Java Essentials:** Elements of Java Program, Java API, Variables and Literals, Primitive Data Types, The String class, Variables, Constants, Operators, Scope of Variables & Blocks, Types of Comment in Java. [C2]	10
Unit-II **Control Statements:** Decision making statements (if, if-else, nested if, else if ladder, switch, conditional operator), Looping statements (while, do-while, for, nested loops), Jumping statements (Break and Continue). [CO1] **Classes and Objects:** Basic concepts of OOPS, Classes and Objects, Modifiers, Passing arguments, Constructors, Overloaded Constructors, Overloaded Operators, Static Class Members, Garbage Collection. [CO1] **Inheritance:** Basics of inheritance, Inheriting and Overriding Superclass methods, Calling Superclass Constructor, Polymorphism, Abstract Classes, Final Class. [CO1]	12
Unit-III **Arrays and Strings:** Introduction to array, Processing Array Contents, Passing array as argument, Returning array from methods, Array of objects, 2D arrays, Array with three or more dimensions. String class, string concatenation, Comparing strings, Substring, Difference between String and String Buffer class, String Tokenizer class. **Interface and Packages:** Basics of interface, Multiple Interfaces, Multiple Inheritance Using Interface, Multilevel Interface, Packages, Create and Access Packages, Static Import and Package Class, Access Specifiers.[CO2]	10

Exception Handling: Introduction, Try and Catch Blocks, Multiple Catch, Nested Try, Finally, Throw Statement, Built-In Exceptions.[CO5]	
Unit-IV **Multithreading:** Introduction, Threads in Java, Thread Creation, Lifecycle of Thread, Joining a Thread, Thread Scheduler, Thread Priority, Thread Synchronization.[CO4] **Applets:** Introduction, Applet Class, Applet Life Cycle, Graphics in Applet, Event-Handling.[CO4] **File and I/O Streams:** File Class, Streams, Byte Streams, Filtered Byte Streams, Random Access File Class, Character Streams.[CO4]	12

Text Books:

1. Programming with Java A Primer, 5th Edition, E. Balagurusamy, TMH.
2. Java Programming for Core and Advanced Learners, Sagayaraja, Denis, Karthik, Gajalakshmi, Universities Press.
3. Java Fundamentals, A Comprehensive Introduction, H. Schildt, D. Skrien, TMH.

Reference Books:

1. Java, The complete Reference, H. Schildt, 7th Edition, TMH.

Course Code: UGCA1938
Course Name: Programming in Java Laboratory

Program: BCA	**L**: 0 **T**: 0 **P**:4
Branch: Computer Applications	**Credits**: 2
Semester: 5th	**Contact hours**: 4 hours per week
Theory/Practical: Practical	**Percentage of numerical/design problems:** 90%
Internal max. marks: 60	**Duration of end semester exam (ESE)**: 3hrs
External max. marks: 40	**Elective status:** Core
Total marks: 100	

Prerequisite: - Basic knowledge of Programming language like Programming in C.
Co requisite: - Knowledge of Object Oriented Concepts through any language like C++.
Additional material required in ESE: - Minor Project.

Course Outcomes: Students will be able to

CO#	Course Outcomes
CO1	Execute Core Java concepts.